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TOWARDS EVENT-DRIVEN RADARS: SPECTRAL SUPER-RESOLUTION AND HARDWARE

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ABSTRACT

Radars predominantly utilize synchronous paradigms to digitize and process waveforms reflected from targets. A common signal model considered in radars is a sum-of-sinusoids, well studied in the field of spectral estimation. In practice, analog-to-digital converters (ADCs) achieve digitization by quantizing both time and amplitude and are fundamentally limited by the bit-budget. This has led to the emergence of low-resolution and 1-Bit ADCs. Here, we propose a different 1-Bit radar architecture that is based on an event-driven philosophy, which allows for exact amplitude recovery. This approach is distinct from previous works as it relies on asynchronous sampling and clockless ADCs. Non-uniform structure of the recorded "events" fundamentally breaks conventional filtering. Event-driven frequency estimation (ED-NFrest) leverages the sinusoidal structure to retrieve parameters directly from triggers. We demonstrate that ED-NFrest achieves bandwidth extension and frequency resolution improvement through hardware experiments with a radar setup.

***Index Terms*—** Event-Driven, Non-Uniform Spectral Estimation, Non-Linear Reconstruction, Radar, Time-Encoded Sampling.

1. INTRODUCTION

Radar systems digitize waveforms reflected from targets by employing analog-to-digital converters (ADCs) designed in accordance with the Shannon–Nyquist paradigm [1]. Among various signal models, the sum-of-sinusoids (SoS) representation is fundamental to both FMCW and Doppler setups [2]. However, the circuit complexity and power consumption of ADCs scale exponentially with the quantization bit budget [3], motivating extensive research on low-resolution schemes [4]. A widely studied example is 1-bit quantization, including signum quantization for direction-of-arrival (DoA) estimation [5] and 1-bit sigma–delta modulation [6]. In radar applications, signum quantizers have been explored with time-varying thresholds [7], additive dithering [8], and multiplicative dithering [9]. While effective to some extent, these approaches compensate for coarse quantization mostly through substantial oversampling.

Event-Driven Sampling. The fundamental bottleneck of the previously mentioned 1-bit quantization schemes is the *lossy* representation, limiting the full potential of radar applications. Shifting from synchronous to asynchronous sampling paradigms, a compelling alternative is Event-Driven Sampling (EDS) that offers simultaneous 1-bit and lossless representation [10]. At the heart of EDS is Lebesgue sampling—only capture information when an *event* occurs, yielding a sequence of non-uniform time stamps. Akin to Shannon's framework, EDS offers an exact discrete representation utilizing an asynchronous architecture, which has been explored in various contexts such as EVENT-DRIVEN [11], TIME-ENCODED [10], IRREGULAR [12], ASYNCHRONOUS [13] or SEND-ON-DELTA [14] sampling. In contrast to the 1-bit synchronous quantization schemes, EDS offers the advantages of perfect signal reconstruction, which unlocks more capabilities of radar systems. Despite these compelling merits, the event-driven radar has not been explored previously. This can be mainly attributed to: (i) At the acquisition front, EDS is an asynchronous scheme for which its hardware investigation has only recently been reported in [15, 16, 17, 18]. This creates difficulties for the subsequent integration with existing radar systems based on a synchronous setup. (ii) At the reconstruction front, existing radar methods are established based on uniform sampling, which does not apply to EDS. This motivates the development of an end-to-end Event-Driven Radar (EDS-Radar) pipeline, which enables low-resolution sampling and high-precision reconstruction.

Challenges and Related Works. With the SoS model, the EDS-Radar signal reconstruction amounts to *event-driven spectral estimation*. Taking SoS as a special case of bandlimited signals, the spectral parameters can be found by applying a sequential reconstruction method that utilizes a time-decoding machine (TDM) and subsequent frequency estimation. Since the parametric structure is not fully leveraged in TDM, such approach exhibits suboptimal performance on estimation resolution. In our previous art [15, 16], we formulate event-driven spectral estimation as non-uniform parameter estimation, where we demonstrate performance enhancement on estimation resolution over the Time-Decoding Machine (TDM) [10] recovery. In [16], we demonstrate super-resolution down to 2 Hz scale of 100 Hz range input SoS signal. Given a fixed time quantization on trigger events (see Section VI in [10]), the maximum input bandwidth is restricted (100 Hz) due to the LASSO formulation (see P1 in Section III in [16]), which does not fulfill the bandwidth requirements of practical radar system (> 1 kHz). While, in [15], an on-grid approach was proposed with better robustness towards time quantization, but it lacks super-resolution capability. The combination of simultaneous high bandwidth and super-resolution necessitates robust non-uniform spectral estimation method, particularly in practical EDS-Radar configurations where the trigger time events are contaminated by clutter and system noise.

Contributions. In this paper, we propose a non-trivial algorithmic hybridization that integrates the ℓ_0-minimization in [15] and off-the-grid spectral estimation in [16], enabling significant performance enhancement in both input bandwidth and frequency resolution.

- We implement Event-Driven Radar by incorporating a custom-built ED-ADC into the acquisition pipeline of the radar system.
- We design a new ED spectral estimation method, attaining 3.1 kHz bandwidth beyond the capability of the state-of-the-art (0.1 kHz).

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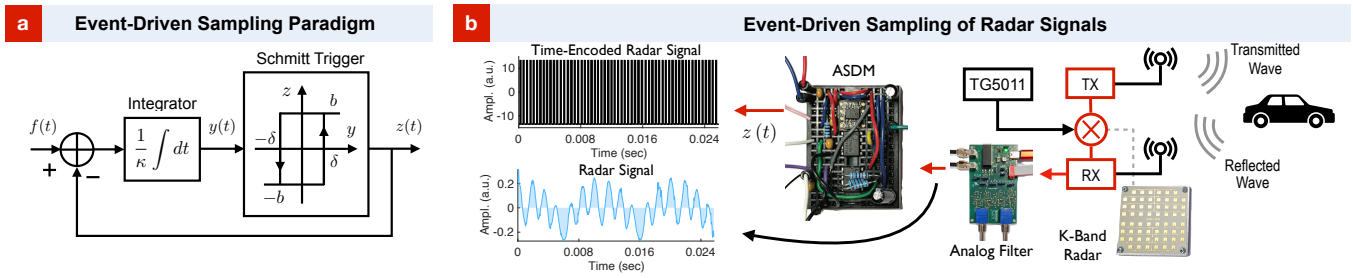


Fig. 1: ED Sampling. (a) Event-Driven Sampler (EDS). (b) Event-Driven Radar (EDS-Radar). We use a K-Band radar with frequency $f_{\mathsf{TX}} = 24.125$ GHz for both FMCW (Fig. 3) and Doppler (Fig. 4) configurations. The analog radar signal is filtered and fed into an Asynchronous Sigma Delta Modulator (ASDM).

- We validate the proposed method through numerical and hardware experiments on both FMCW and Doppler configurations, enabling super-resolved spectral estimation down to 0.5 Hz of 1.8 kHz.

Notations. We denote integers, reals and complex numbers as $\mathbb{Z}$, $\mathbb{R}$ and $\mathbb{C}$, respectively. A set of N contiguous integers is denoted as $\mathbb{I}_N = \{0, \dots, N-1\}, N \in \mathbb{Z}^+$. The indicator function on the domain $\mathcal{D}$ is denoted as $\mathbb{1}_{\mathcal{D}}$. Continuous functions are represented by $f(t), (t) \in \mathbb{R}$ and their discrete counterparts are written as $f[n] = f(t)|_{t=nT}, n \in \mathbb{Z}$ where $T = {}^{2\pi}/_{\omega_s} > 0$ is the sampling period. Vectors and matrices are written in bold lowercase and uppercase fonts, *e.g.* $\mathbf{f} \in \mathbb{R}^N$ and $\mathbf{F} \in \mathbb{R}^{N \times M}$. We define inner product as $\langle \cdot, \cdot \rangle$. The max-norm of a function is defined as $\|f\|_\infty = \inf\{c_0 \geqslant 0 : |f(t)| \leqslant c_0\}$; for sequences, we use, $\|f\|_\infty = \max_n |f[n]|$. The ℓ_2-norm of a sequence is defined as $\|f\|_2 = \sqrt{\sum_{n=0}^{N-1} |f[n]|^2}$. We denote the mean-square error (MSE) between $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$ as $\mathcal{E}_2(\mathbf{x}, \mathbf{y}) \stackrel{\text{def}}{=} \frac{1}{N} \sum_{n=0}^{N-1} |x[n] - y[n]|^2$.

2. EVENT-DRIVEN SAMPLING

In this paper, we consider the acquisition of radar signals with an event-driven approach summarized in Fig. 1. Event-Driven Sampler (EDS) encodes amplitude information of a signal $f(t)$ as a time stamp sequence $\{t_n\}_{n \in \mathbb{Z}}$ provided that $|f(t)| \leqslant c < b$. Bounded signal $f(t)$ is biased by the output of the Schmitt Trigger $z(t)$ with two modes of operation. In the first mode, $z(t) = -b$ resulting in the increase of the output of the integrator $y(t)$ from $-\delta$ to δ. When $y(t)$ reaches δ, Schmitt Trigger flips to $z(t) = b$ making $y(t)$ to decrease from δ to $-\delta$. When $y(t)$ reaches $-\delta$, Schmitt Trigger transitions back to the first state $z(t) = -b$. The negative feedback ensures $y(t)$ is within the range from $-\delta$ to δ. Non-uniform time transitions of $z(t)$ depend on signal $f(t)$ and EDS design parameters δ, κ and b. Modulus of the output $|z(t)| = b$ is constant.

Signal Model. We assume $f(t)$ is a sum of sinusoids (SoS)

$$f(t) = \sum_{k=0}^{K-1} c_k e^{\jmath \omega_k t}, \ \omega_k = 2\pi f_k, \ \Omega = \max_k |\omega_k| \qquad (1)$$

which creates time stamps $\{t_n\}_{n \in \mathbb{Z}}$ and $\omega_k, k \in \mathbb{I}_K$ are frequencies.

Problem Statement. Given *noisy* non-uniform time stamps $\{t_n\}_{n \in \mathbb{Z}}$, estimate sinusoidal parameters $\{c_k, \omega_k\}_{k \in \mathbb{I}_K}$.

Sequential Spectral Estimation using Time-Decoding. Since the input $f(t)$ is Ω-bandlimited, a potential solution is to use sequential reconstruction – recovery via time-decoding machine (TDM) [10, 11] followed by spectral estimation [19, 20, 21].

Step #1: Recovery of $f(t)$ via TDM. As shown in Fig. 1(a), the sequence of trigger times $\{t_n\}_{n \in \mathbb{Z}}$ is generated as

$$\int_{t_n}^{t_{n+1}} f(t)\, dt = (-1)^n \left[-b(t_{n+1} - t_n) + 2\kappa\delta\right]. \qquad (2)$$

A simplified version of the EDS is considered, and without a loss of generality, assuming $|f(t)| \leqslant c < b = 1$ and $\kappa = 1/2$,

$$\int_{t_n}^{t_{n+1}} f(t)\, dt = (-1)^n \left[\delta - t_{n+1} + t_n\right]. \qquad (3)$$

Using (3), we bound the distance between t_n and t_{n+1}.

Lemma 1. *For an arbitrary signal $f(t), |f(t)| \leqslant c < 1$, the distance between consecutive time triggers t_n and t_{n+1} is bounded as*

$$\frac{\delta}{1+c} \leqslant t_{n+1} - t_n \leqslant \frac{\delta}{1-c}. \qquad (4)$$

Proof. With $|f(t)| \leqslant c < 1$, we know

$$-c(t_{n+1} - t_n) \leqslant \int_{t_n}^{t_{n+1}} f(t)\, dt \leqslant c(t_{n+1} - t_n). \qquad (5)$$

By substituting for the integral from (3) and solving for $t_{n+1} - t_n$, we obtain the bounds. □

Recovery of $f(t)$ is achieved via a sequence of recursive functions. Denote $f = f(t)$ and define the operator $\mathcal{A}$ as

$$\mathcal{A}f = \sum\nolimits_{n \in \mathbb{Z}} f(s_n)\, g_n(t), \ g_n(t) \stackrel{\text{def}}{=} \left(h_\Omega * \mathbb{1}_{[t_{n+1}, t_n]}\right)(t)$$

where $s_n = (t_n + t_{n+1})/2$, and $h_\Omega(t) = \sin(\Omega t)/\pi t$. Let $f_m(t)$ be a recursive sequence of bandlimited functions defined as

$$f_{m+1} = f_m + \mathcal{A}(f - f_m), \ m \in \mathbb{N} \qquad (6)$$

with the initial condition $f_0 = \mathcal{A}f$. We can recover $f(t)$ from $\{t_n\}_{n \in \mathbb{Z}}$ given that $\lim_{m \to \infty} f_m(t) = f(t)$ and $\delta < \frac{(1-c)\pi}{\Omega}$.

Step #2: Spectral Estimation via Prony's Method. Given the signal recovery $f(t)$, $\{\omega_k\}_{k \in \mathbb{I}_K}$ can be retrieved from uniform samples $\{f(mT)\}_{m \in \mathbb{I}_M}$ using Prony's method [22] given that $T < \pi/\Omega$ and $M \geqslant 2K$. More robust frequency estimates can be obtained with matrix pencil [20] and model-fitting [23]. With $\{\omega_k\}_{k \in \mathbb{I}_K}$ known, the amplitudes $\{c_k\}_{k \in \mathbb{I}_K}$ can be estimated via least-squares.

Remarks. Sequential recovery has the following limitations:

- TDM relies only on the *bandlimited constraint* without considering the *sinusoidal structure* of the signal, and this is likely to compromise the robustness of spectral estimation.

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- Time quantization negatively impacts bandlimited reconstruction of TDM leading to suboptimal performance of spectral estimation.

3. EVENT-DRIVEN FREQUENCY ESTIMATION

We leverage the structure of $f(t)$ and use a model-fitting method proposed in [16] to estimate parameters from the event-driven sequence $y[n]$. For $f(t)$ with a structure in (1), we express $y[n]$ as

$$y[n] = \int_{t_n}^{t_{n+1}} f(t)\,dt \overset{(1)}{=} \sum\nolimits_{k=0}^{K-1} \underline{c}_k \left(e^{\jmath\omega_k t_{n+1}} - e^{\jmath\omega_k t_n}\right) \quad (7)$$

where $\underline{c}_k = \frac{c_k}{\jmath\omega_k}$. We calculate $\{y\}_{n\in\mathbb{I}_N}$ from $\{t_n\}_{n\in\mathbb{I}_{N+1}}$ as

$$y[n] \overset{(3)}{=} (-1)^n \left[\delta - (t_{n+1} - t_n)\right] \quad (8)$$

which implicitly encodes signal parameters $\{c_k, \omega_k\}_{k\in\mathbb{I}_K}$.

ED Non-Uniform Spectral Estimation. (8) is true only in noiseless scenarios. In practice, $y[n]$ is corrupted by noise

$$y[n] = \sum\nolimits_{k=0}^{K-1} \underline{c}_k \left(e^{\jmath\omega_k t_{n+1}} - e^{\jmath\omega_k t_n}\right) + \eta[n], \quad n \in \mathbb{I}_N \quad (9)$$

where $\eta[n]$ models noise due to hardware imperfections and time jitter. Given (9), the spectral estimation [15] can be posed as

$$\mathcal{J}(\underline{c}_k, \omega_k) = \sum_{n=0}^{N-1} \left| y[n] - \sum_{k=0}^{K-1} \underline{c}_k \left(e^{\jmath\omega_k t_{n+1}} - e^{\jmath\omega_k t_n}\right) \right|^2. \quad (10)$$

The non-uniform nature of $y[n]$ makes the optimization problem in (10) challenging. We utilize a *divide-and-conquer* strategy where we use on-grid approximation followed by resolution enhancement.

Step #1: On-Grid Approximation. The objective function in (10) is non-linear and non-convex. Our first step is to relax (10) as a series of 1D optimizations. We utilize an on-grid approximation to solve the problem in an orthogonal-matching pursuit (OMP) manner, widely used in uniform spectral estimation [24, 25] together with NOMP [26]. Let $\mathbf{p}(\omega) \in \mathbb{C}^N$ where $[\mathbf{p}(\omega)]_n = e^{\jmath\omega t_{n+1}} - e^{\jmath\omega t_n}$, the minization on $\mathcal{J}$ in (10) results in the following optimization

$$\min_{\mathbf{c}} \|\mathbf{c}\|_{\ell_0}, \quad \text{subject to} \quad \mathbf{P}_L\mathbf{c} = \mathbf{y} \quad (11)$$

where $\mathbf{P}_L = [\mathbf{p}(\omega_0), \ldots, \mathbf{p}(\omega_I)]$ and $\omega_l = \frac{2\Omega l - \Omega(L+1)}{L+1}, l \in \mathbb{I}_{L+1}$. Frequencies $\tilde{\omega}_k$ are found iteratively. We start by denoting

$$\mathbf{y}_m = \mathbf{y} - \sum\nolimits_{k\neq m} \tilde{c}_k \mathbf{p}(\tilde{\omega}_k) \quad (12)$$

where $\{\tilde{c}_k, \tilde{\omega}_k\}_{k=0,k\neq m}^{K-1}$ are known. Then (10) becomes

$$\mathcal{J}(c_m, \omega_m) = \|\mathbf{y} - c_m\mathbf{p}(\omega_m)\|_2^2. \quad (13)$$

Minimizing (13) on c_m and ω_m can be decoupled. Amplitude estimate $\tilde{c}_m$ can be characterized by the frequency estimate $\hat{\omega}_m$ as $\tilde{c}_m = \langle \mathbf{p}(\tilde{\omega}_m), \mathbf{y}_m\rangle / \|\mathbf{p}(\tilde{\omega}_m)\|_2^2$. With $\tilde{c}_m$, we calculate $\tilde{\omega}_m$ as

$$\tilde{\omega}_m = \arg\max_{\tilde{\omega}_m} |\langle \mathbf{u}(\tilde{\omega}_m), \mathbf{y}_m\rangle| \quad (14)$$

where $\mathbf{u}(\tilde{\omega}_m) = \frac{\mathbf{p}(\tilde{\omega}_m)}{\|\mathbf{p}(\tilde{\omega}_m)\|_2}$. The estimate $\tilde{\omega}_m$ is obtained as the location of the dominant peak of the magnitude of the cross-correlation function $\mathcal{C}(\omega) = |\langle \mathbf{u}(\omega), \mathbf{y}_m\rangle|$. $\mathcal{C}(\omega)$ has K local maxima corresponding to frequencies $\{\omega_k\}_{k\in\mathbb{I}_K}$. In case K is unknown, the number of targets can be obtained by thresholding dominant peaks.

Step #2: Enhancing Resolution. We further refine on-grid estimates with perturbations. Given that I is large, the perturbation $\Delta\omega_k$ is sufficiently small and (9) can be rewritten as

$$y[n] = \sum\nolimits_{k=0}^{K-1} \left(\jmath e^{\jmath\tilde{\omega}_k t_{n+1}} t_{n+1} - \jmath e^{\jmath\tilde{\omega}_k t_n} t_n\right) \underline{c}_k \Delta\omega_k + \sum\nolimits_{k=0}^{K-1} \left(e^{\jmath\tilde{\omega}_k t_{n+1}} - e^{\jmath\tilde{\omega}_k t_n}\right) \underline{c}_k + o(\Delta\omega_k) + \eta[n].$$

Let $\mathbf{q}(\omega) \in \mathbb{C}^N$, $[\mathbf{q}(\omega)]_n = \jmath e^{\jmath\omega t_{n+1}} t_{n+1} - \jmath e^{\jmath\omega t_n} t_n$. The perturbations can be found by solving the following minimization problem

$$\min_{\mathbf{c},\mathbf{r}} \|\mathbf{y} - \mathbf{P}_K\mathbf{c} - \mathbf{Q}\mathbf{r}\|_2^2 \quad (15)$$

where $\mathbf{P}_K = [\mathbf{p}(\omega_0), \ldots, \mathbf{p}(\omega_{K-1})]$, $\mathbf{Q} = [\mathbf{q}(\omega_0), \ldots, \mathbf{q}(\omega_{K-1})]$ and $\mathbf{r} = \mathbf{c} \odot \boldsymbol{\Delta}\omega$. (15) is a least-squares problem with a solution

$$\left[\mathbf{c}^\top, \mathbf{r}^\top\right]^\top = \left([\mathbf{P}_K, \mathbf{Q}]^{\mathrm{H}}[\mathbf{P}_K, \mathbf{Q}]\right)^{-1} [\mathbf{P}_K, \mathbf{Q}]^{\mathrm{H}}\mathbf{y}. \quad (16)$$

Given $\boldsymbol{\Delta}\omega$, we update estimates via $\tilde{\omega}_k \leftarrow \omega_k + \mathsf{Re}(\boldsymbol{\Delta}\omega),\ k \in \mathbb{I}_K$.

Convergence. The proposed non-uniform spectral estimation algorithm (ED-NFrest) is bound to converge to at least some local minimum [27], because the minimization in (10) results in

$$\mathcal{J}\left(\tilde{\underline{\mathbf{c}}}_k^{[m+1]}, \tilde{\boldsymbol{\omega}}_k^{[m+1]}\right) \leqslant \mathcal{J}\left(\tilde{\underline{\mathbf{c}}}_k^{[m]}, \tilde{\boldsymbol{\omega}}_k^{[m]}\right). \quad (17)$$

ED-NFrest converges in a few steps if a separation between frequencies is sufficiently large. We refine $\{\omega_k\}_{k\in\mathbb{I}_K}$ for a fixed number of iterations, or until the change of $\mathcal{J}(\underline{c}_k, \omega_k)$ between consecutive iterations is small. The algorithm is summarized in Algorithm 1.

Event-Driven Non-Uniform Frequency Estimation (**ED-NFrest**)

Input: Time sequence $\{t_n\}_{n=0}^N$.
1: Compute the samples $\{y[n]\}_{n=0}^{N-1}$ using (8).
2: Compute $\{\tilde{\underline{\mathbf{c}}}_k^{[0]}, \tilde{\boldsymbol{\omega}}_k^{[0]}\}_{k=0}^{K-1}$ by solving (11).
3: **for** $m = 1$ to `max. iterations` **do**
4: Update $\tilde{\underline{\mathbf{c}}}_k^{[m]}, \tilde{\boldsymbol{\omega}}_k^{[m]}$ via solving (15).
5: **end for**
Output: Signal parameters $\{c_k, \omega_k\}_{k\in\mathbb{I}_K}$.

4. NUMERICAL EXPERIMENTS

To validate the robustness of ED-NFrest to time quantization, we run simulations with a signal with frequencies $[f - 9.75, f, f + 9.75]$ Hz to test the frequency resolution limit. Input $f(t)$ is bounded such that $|f(t)| \leqslant 0.65 < b$. We use EDS with parameters of ASDM hardware ($b = 2.14$, $\kappa = 2.51 \times 10^{-4}$ and $\delta = 0.96$). The bandwidth of TDM is $f_{\mathsf{max}} = \frac{(b-c)}{4\kappa\delta} = 1552$ Hz. Comparison between the recovery with TDM and ED-NFrest is depicted in Fig. 2. ED-NFrest with $f_s = 10$ kHz achieves 7.4 dB reconstruction improvement compared to TDM with $f_s = 25$ kHz and 2.4 dB improvement compared to TDM with $f_s = 50$ kHz.

5. HARDWARE EXPERIMENTS

To validate ED-NFrest in practice, we conduct hardware experiments with a signal generator of `3024A` and a radar setup depicted in Fig. 1 (b). We utilise `PicoScope 3403D` to digitize both the ground truth

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Table 1: Hardware Experimental Parameters and Performance Evaluation.

Figure	Exp. No.	N	f_s (kHz)		κ	δ	b	$\|g\|_\infty$	f_k	$\widetilde{f}_k$ (Hz)		$\mathcal{E}_2\left(\mathbf{f}_k, \widetilde{\mathbf{f}}_k\right)$		$\mathcal{E}(\widetilde{\mathbf{g}}, \mathbf{g})$	
			(ED-NFrest)	(TDM)					(Hz)	ED-NFrest	TDM	ED-NFrest	TDM	ED-NFrest	TDM
Amplitude Modulation															
—	1	1383	8.00	50.00	2.51×10^{-4}	0.96	2.14	0.48	[495.0, 500.0, 505.0]	[494.8, 499.9, 505.0]	[494.4, 499.7, 504.7]	$\mathbf{1.93 \times 10^{-2}}$	1.43×10^{-1}	$\mathbf{9.04 \times 10^{-4}}$	3.33×10^{-3}
—	2	806	50.00	100.00	2.50×10^{-4}	0.96	2.14	0.47	[995.0, 1000.0, 1005.1]	[994.3, 999.8, 1005.1]	[993.2, 1001.7, 1002.4]	$\mathbf{1.88 \times 10^{-1}}$	4.44×10^{0}	$\mathbf{1.00 \times 10^{-3}}$	2.41×10^{-3}
—	3	895	100.00	100.00	2.51×10^{-4}	0.96	2.15	0.48	[2195.0, 2200.1, 2205.0]	[2195.2, 2200.8, 2205.0]	[2110.9, 2196.5, 2203.4]	$\mathbf{1.83 \times 10^{-1}}$	2.36×10^{3}	$\mathbf{3.44 \times 10^{-3}}$	1.05×10^{-2}
FMCW Radar for Ranging															
Fig. 3	4	4363	250.00	250.00	1.64×10^{-4}	0.95	13.60	0.64	[30.0, 60.0, 120.0]	[33.0, 60.5, 120.4]	[34.9, 61.8, 119.8]	$\mathbf{3.11 \times 10^{0}}$	9.13×10^{0}	$\mathbf{7.78 \times 10^{-3}}$	3.33×10^{-2}
—	5	1604	500.00	500.00	1.60×10^{-4}	0.95	13.55	0.34	[100.0, 600.0]	[99.6, 599.1]	[101.2, 596.0]	$\mathbf{4.61 \times 10^{-1}}$	8.77×10^{0}	$\mathbf{3.14 \times 10^{-3}}$	1.06×10^{-2}
Doppler Radar for Speed Estimation															
—	6	3391	10.00	100.00	1.82×10^{-4}	0.95	2.13	1.87	[84.0, 447.0]	[87.5, 447.4]	[54.5, 447.0]	$\mathbf{6.30 \times 10^{0}}$	4.34×10^{2}	$\mathbf{5.42 \times 10^{-3}}$	6.28×10^{-3}
—	7	396	100.00	100.00	1.80×10^{-3}	0.95	13.62	0.73	[1788.0]	[1788.4]	[1789.5]	$\mathbf{1.99 \times 10^{-1}}$	2.18×10^{0}	$\mathbf{2.70 \times 10^{-4}}$	1.95×10^{-1}
Fig. 4	8	63	500.00	500.00	1.81×10^{-3}	0.95	13.64	0.88	[3129.0]	[3131.1]	[1374.6]	$\mathbf{4.36 \times 10^{0}}$	3.08×10^{6}	$\mathbf{9.38 \times 10^{-3}}$	1.74×10^{-1}

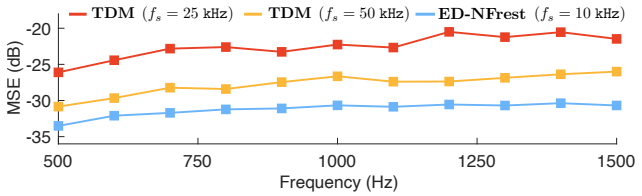


Fig. 2: Numerical experiment with EDS ($b = 2.14$, $\kappa = 2.51 \times 10^{-4}$, $d = 0.96$). Signal for each point contains frequencies $[f - 9.75, f, f + 9.75]$ Hz.

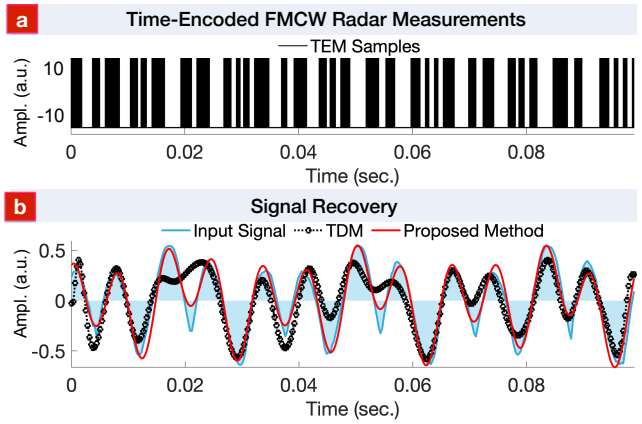


Fig. 3: Hardware Experiments with FMCW Radar with bandwidth $f_{\max} = 120$ Hz. ED samples are captured with $f_s = 250$ kHz. Frequency resolution for TDM is 4.9 Hz, while ED-NFrest achieves resolution of 3 Hz. Hardware imperfections cause sharp transitions that negatively impact reconstruction.

and output of the ASDM. The radar signal is characterized as a sum of sinusoids where the frequency contains key information such as *range* for FMCW radar and *speed* for Doppler [2]. We demonstrate the capabilities of ED-NFrest by quantitatively pushing performance metrics (i) bandwidth, (ii) frequency resolution, and (iii) reconstruction MSE, and benchmark against sequential recovery.

Amplitude Modulation (AM). In Exp. 1 in Table 1, ED-NFrest achieves super-resolution up to 0.2 Hz with a bandwidth of 505 Hz and $f_s = 8$ kHz while reducing oversampling factor 6.25× compared to TDM. In Exp. 3, we push the TDM to its limits such that its bandwidth $f_{\max} = \frac{(b-c)}{4\kappa\delta} = 1733$ Hz is exceeded. TDM reconstruction results in abrupt changes of the signal, which negatively impact spectral estimation. In these challenging conditions, ED-NFrest achieves frequency resolution up to 0.8 Hz.

FMCW Radar. For FMCW radar, we modulate the instantaneous frequency with TG5011. Given a beat frequency ω_k, the range of a target can be calculated as $R_k = \frac{cT_{\mathsf{M}}}{8\pi f_{\mathsf{M}}}\omega_k$ where $c = 3 \times 10^8$ m/s is the speed of light, $f_{\mathsf{M}} = 145$ MHz is the modulation depth and T_{M} is the chirp period ($T_{\mathsf{M}} = 33$ ms for Exp. 4 and $T_{\mathsf{M}} =$ ms for Exp. 5). For Exp. 4 depicted in Fig. 3, we capture $N = 4363$ ED samples with a frequency $f_s = 250$ kHz. Demodulated signal exhibits sharp transitions due to hardware non-idealities and periodic harmonics. ED-NFrest achieves resolution up to 3 Hz and the reconstruction improvement allows for better target cross-section estimation.

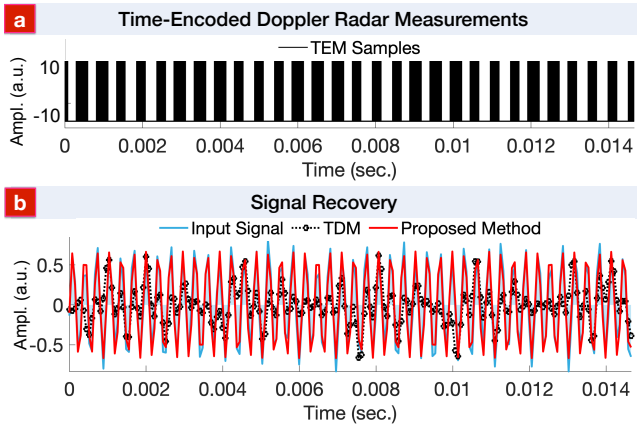


Fig. 4: Hardware Experiments with Doppler Radar. ED samples are captured with $f_s = 500$ kHz. ED-NFrest achieves reconstructions of a signal with a frequency $f_k = 3129$ Hz (target with $v = 70$ km/h) with a resolution of 2.1 Hz. Due to high bandwidth and effects of time quantization, TDM fails.

Doppler Radar. Given a frequency ω_k, the speed of the target can be calculated as $v_k = \frac{c}{4\pi f_{\mathsf{TX}}}\omega_k$ where f_{TX} is the transmitted frequency and $c = 3 \times 10^8$ m/s is the speed of light. In Exp. 6 in Table 1, the first target is a walking person, and the second is emulated with DT-1 Target Simulator. In subsequent experiments, we use only DT-1. With increasing bandwidth, TDM reconstruction quality decreases as shown in Exps. 6 to 8. In Exp. 8, we set the speed on DT-1 to $v_k = 70 \pm 1$km/h. ED measurements sampled with a frequency $f_s = 500$ kHz and corresponding signal reconstruction are depicted in Fig. 4. Due to high bandwidth and time quantization, TDM fails. ED-NFrest achieves frequency resolution within 2.1 Hz resulting in 1.5×10^{-2} km/h speed resolution compared to the received signal.

6. CONCLUSION

Event-Driven Sampling converts amplitude information into a sequence of non-uniform time triggers. In this paper, we integrate Event-Driven Sampler into the acquisition pipeline of the radar system. We show ED-NFrest recovers signals with higher frequencies than sequential recovery. In practice, this allows for estimation of faster targets with Doppler radar. Future work includes investigating the impact of ASDM parameters on the performance of ED-NFrest.

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